Evaluation

Our agent builds an estimate from a project description using similar bids, and we think it'll be able to aggregate relevant historical bids easier and faster than doing it manually. To evaluate this, we created a table of 10 different locations (using coordinates) and project titles and manually gathered 2-3 bids that were most relevant to evaluating the price of bid items, particularly loosely-scoped items. We are aiming for an 80% "hit" rate of relevant bids.

"Relevance" prioritized:

1. Project type (similar projects or projects with the same task are prioritized)
2. Distance/Cluster, if there are sufficient bid items
    - Region, if there are not (South / North California)
3. Recency

** It should be noted that there is some subjectivity in what these "ground truth" bids should be. In addition, upon further research, recency should've been given higher weight than it was in this assessment.

The AI had two trials where it would attempt to produce the bids it believes to be the most relevant. The AI has the same number of "guesses" as the number of correct bids. To help in the evaluation of the AI, we also included a "simplest" evaluation. The simplest methodology simply consisted of using a keyword search sorted by date, not considering location.

Failures were put into the following categories:

- Gathered a superficially similar but irrelevant bid
- Bid is too old
- Bid is irrelevant
- Bid is less relevant than others in terms of proximity/items

"Superficially similar but irrelevant" is considered the failure condition of our AI. The major concern is that it would provide a bid that'll misinform an estimator, which is worse than the bid being merely less relevant.

Scores:

- AI #1 - Most relevant: 11/24, Failed: 1/24
- AI #2 - Most Relevant: 12/24, Failed: 2/24
- Simplest: 6/24

Conclusion:

Despite underperforming our success metric significantly, our AI worked better in terms of raw score compared to the "simple" method thanks to being able to compare items and distance, although at the cost of also rarely including old and superficially relevant items.

Team breakdown: Tony - Evaluation, conclusion, AI trials, simple method, scoring. Tony+Calvin -